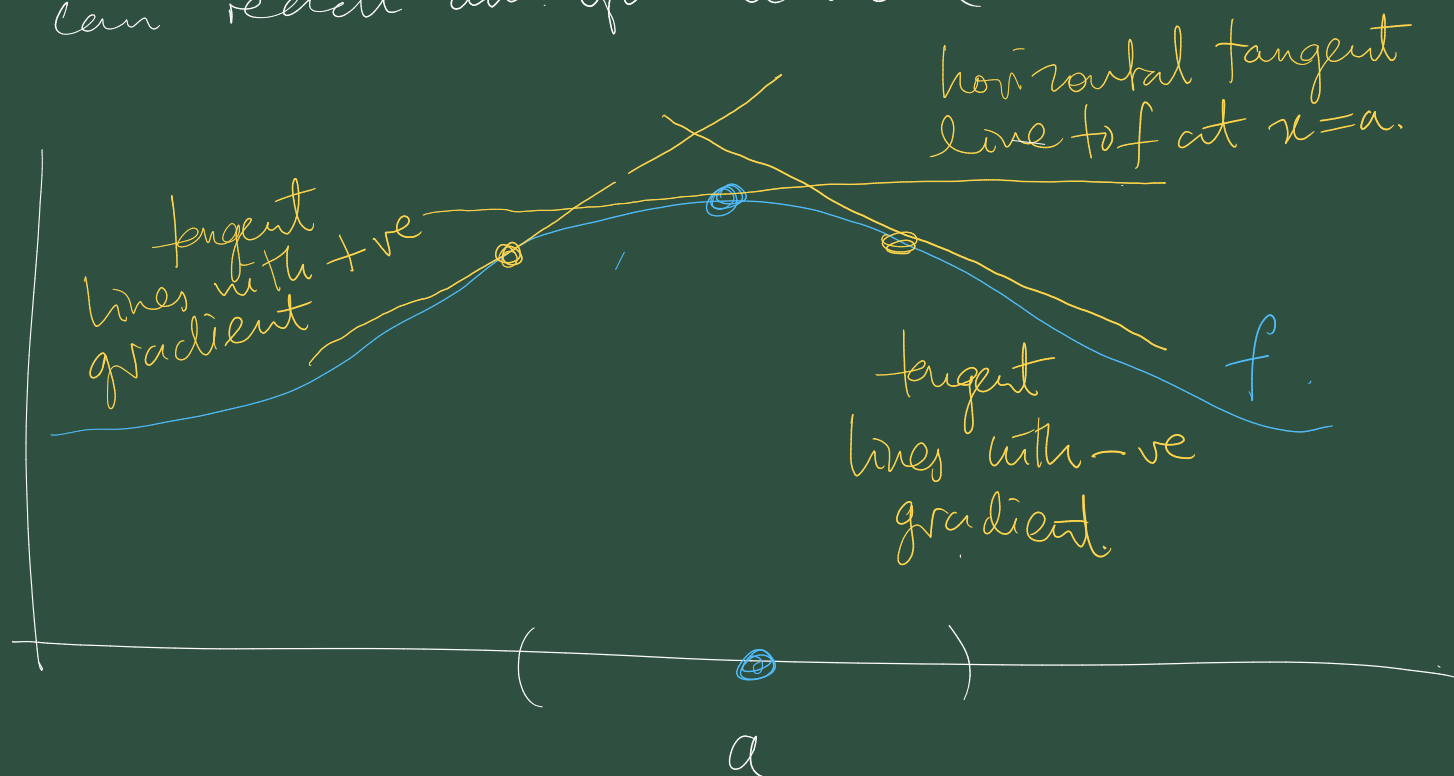


- Applying differentiation to find solutions to optimization problems. (the largest, smallest, cheapest, most expensive, ---).

- Stationary point / critical point.

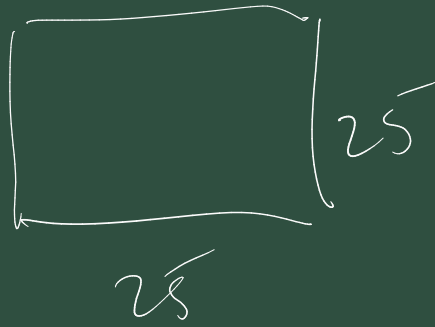
- Consider how a continuous function can reach an optimal value.



When f has a local maximum at $x=a$ when $f'(a) = 0$, and $f'(x) > 0$ for $x < a$, and $f'(x) < 0$ for $x > a$.

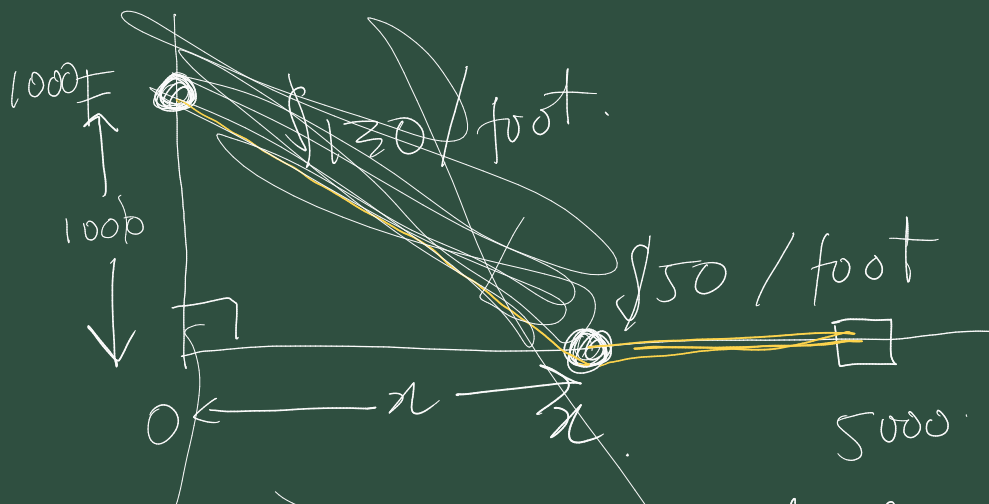
So as x passes through a (from left to right), the derivative $f'(x)$ is decreasing.

the square



With a straightforward perimeter constraint, the rectangle with maximal area, is a square.

Example 4.3.7.



If we build on land from station to x and then under the sea from x to F the cost of this pipeline is

$$C(x) = 50(5000 - x) + 130\sqrt{x^2 + 1000^2}$$

We want to minimize $C(x)$ for x in the range $0 \leq x \leq 5000$

